🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.EE.7

Solve Multiplication and Division Equations

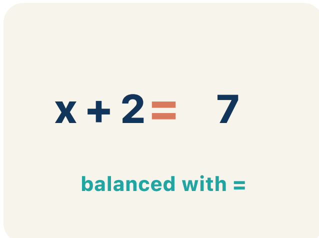
Lesson 7-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary

Level 2 Standard

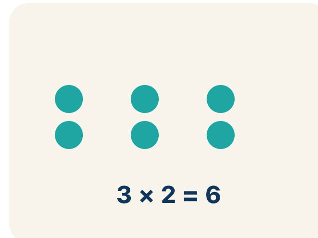
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Multiplication and division are inverses: 3 × 7 = 21, so 21 ÷ 3 = 7

Inverse operation

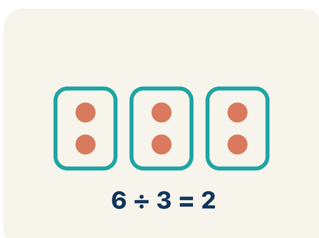
Write the definition:



3 × 7 = 21 means 3 groups of 7 equals 21

Multiply

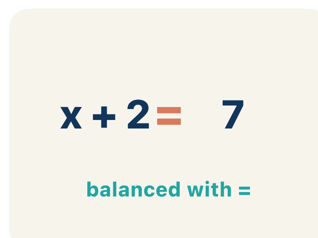
Write the definition:



36 ÷ 4 = 9 means 36 split into 4 equal groups gives 9 per group

Divide

Write the definition:



3x = 21 → divide both sides by 3 → x = 7 (x is alone)

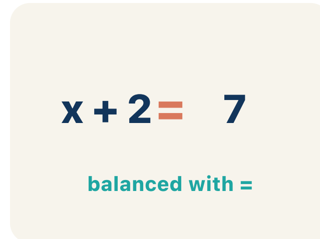
Isolate

Write the definition:

In 6x = 18, the coefficient is 6; divide both sides by 6 to get x = 3.

Coefficient

Write the definition:



x = 3 is the solution of 6x = 18 because 6 × 3 = 18.

Solution

Write the definition:

Guided Notes

Level 2 Standard

WHAT WE'RE LEARNING TODAY

I can solve one-step multiplication and division equations using inverse operations.

Fill in each blank as we go. Use the Word Bank to help you.

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Inverse operation | Multiply | Divide | Isolate | Coefficient | Solution

Tap any word to see what it means and a picture.

1. An operation that undoes another, like division undoing multiplication, is an __________.

2. To undo dividing by a number, I __________ both sides by that number.

3. To undo multiplying by a number, I __________ both sides by that number.

4. To get the variable alone on one side is to __________ it.

5. The number multiplied by a variable is the __________.

6. The value of the variable that makes an equation true is the __________.

Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Solve: 6x = 42

A. x = 7

B. x = 48

C. x = 36

D. x = 252

Step 1 Divide both sides by 6: x = 42 ÷ 6 = 7.

Step 2 Check: 6 × 7 = 42 ✓

✅ **Answer:** A. x = 7

🧩 **Try — put the steps in order**

Drag the cards (or use the ▲▼ buttons) to put the solution steps in the right order, then press **Check**.

Check: 6 × 7 = 42 ✓

Divide both sides by 6: x = 42 ÷ 6 = 7.

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Solve: n / 3 = 8

A. n = 24

B. n = 11

C. n = 5

D. n = 3

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Solve: 9m = 63

A. m = 7

B. m = 54

C. m = 72

D. m = 567

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Vault B is sealed with x / 3 = 6. Crack the lock: what is x?

A. x = 18

B. x = 2

C. x = 9

D. x = 3

Show your work:

2. Vault C is sealed with 5n = 45. Crack the lock: what is n?

A. n = 9

B. n = 225

C. n = 40

D. n = 50

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

Sentence starter: Equation: ____. Solving: ____. The answer ____ is reasonable because ____. A division situation with the same answer: ____.

Show your work:

Reflect — Exit Ticket

Solve: w / 8 = 6

A. w = 48

B. w = 14

C. w = 0.75

D. w = 2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Inverse operation
2. **Notes 2:** Multiply
3. **Notes 3:** Divide
4. **Notes 4:** Isolate
5. **Notes 5:** Coefficient
6. **Notes 6:** Solution
7. **We Try (notes):** A. n = 24 *— Multiply both sides by 3: n = 8 × 3 = 24. Check: 24 / 3 = 8 ✓*
8. **You Try (notes):** A. m = 7 *— Divide both sides by 9: m = 63 ÷ 9 = 7. Check: 9 × 7 = 63 ✓*
9. **Try It 1:** A. x = 18 *— x is divided by 3, so multiply both sides by 3: x = 6 x 3 = 18. Check: 18 / 3 = 6.*
10. **Try It 2:** A. n = 9 *— The 5 is multiplying n, so divide both sides by 5: n = 45 / 5 = 9. Check: 5 x 9 = 45.*
11. **Exit Ticket:** A. w = 48 *— Multiply both sides by 8: w = 6 × 8 = 48. Check: 48 / 8 = 6 ✓*